

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I C. PRIYANKA / (192511074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C. PRIYANKA

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.
During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.
2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to

cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

ANSWERS

1) Root Causes of the Problems

1. Slow Internet Speed

- High number of users connected to the same Access Points (APs).
- Insufficient bandwidth during peak office hours.
- Network congestion caused by cloud applications and video conferencing.

2. Frequent Wi-Fi Disconnections

- Poor roaming configuration between Access Points.
- Improper placement of APs across different floors.
- Weak signal overlap causing roaming delays.

3. **Weak Wireless Signal**

- Physical obstacles such as walls, glass partitions, and furniture.
- Insufficient number of Access Points in conference rooms.
- Radio frequency interference from nearby wireless devices.

4. **Video Conferencing Freezing**

- High network latency and packet loss.
- Limited Quality of Service (QoS) configuration.
- Overloaded wireless channels.

5. **Uneven Traffic Distribution**

- Some Access Points have too many connected devices.
- No load balancing between APs.
- Improper channel and power configuration.

6. **WWAN Connectivity Loss**

- Poor 4G/5G coverage in rural and highway locations.
- Signal blockage due to terrain and distance from cellular towers.
- Network handoff issues while moving.

7. **Security Risks**

- Devices using outdated protocols such as WEP or WPA.
- Weak authentication methods.
- Lack of regular security updates.

Recommended Solutions

WLAN Optimization

- Perform a wireless site survey.
- Install additional Access Points in weak coverage areas.
- Enable seamless roaming using **802.11r**, **802.11k**, and **802.11v**.
- Configure automatic load balancing across Access Points.
- Use dual-band (2.4 GHz and 5 GHz) or Wi-Fi 6/6E Access Points.
- Select non-overlapping wireless channels to reduce interference.
- Implement QoS to prioritize voice and video traffic.

WWAN Optimization

- Use multiple mobile network providers for redundancy.
- Deploy 5G wherever available.
- Install external high-gain antennas for better signal reception.
- Use VPN with automatic reconnect for secure remote access.
- Enable offline data storage with automatic synchronization when connectivity returns.

Security Improvements

- Replace WEP/WPA with **WPA3** encryption.
- Implement **802.1X authentication** with a RADIUS server.
- Use strong passwords and Multi-Factor Authentication (MFA).
- Regularly update firmware of Access Points and wireless devices.
- Monitor the wireless network using Intrusion Detection/Prevention Systems (WIDS/WIPS).

Expected Outcomes

- Faster wireless internet performance.
- Seamless Wi-Fi roaming across all six floors.
- Strong wireless coverage in conference rooms.
- Smooth and uninterrupted video conferencing.
- Balanced network traffic across Access Points.
- Improved WWAN connectivity for field engineers.
- Secure wireless communication with modern encryption.
- Higher productivity and reliable access to cloud applications.

2) Problems Identified (Root Causes)

1. Delay in Online Transactions

- Congestion in ATM Virtual Circuits (VCs).
- High utilization of ATM switches.
- Insufficient bandwidth allocation.

2. Voice Call Jitter and Packet Loss

- Incorrect ATM service category assigned to VoIP traffic.
- Network congestion causing delay variation.
- Cell loss due to overloaded switches.

3. Poor Video Conferencing Quality

- High latency and packet loss.
- Inadequate Quality of Service (QoS).
- Insufficient bandwidth for video traffic.

4. ATM Switch Overload

- Excessive traffic passing through a few switches.
- Uneven traffic distribution.
- Lack of traffic engineering.

5. Virtual Circuit (VC) Congestion

- Too many applications sharing the same VC.
- Improper routing of virtual circuits.
- Limited available bandwidth.

6. Inefficient Bandwidth Allocation

- Bandwidth not prioritized according to application requirements.
- Poor traffic management policies.

7. Incorrect ATM Service Categories

- Real-time applications not using **CBR (Constant Bit Rate)**.
- Variable traffic assigned to inappropriate service classes.
- Inefficient use of ATM resources.

Recommended Solutions

ATM Network Optimization

- Monitor ATM switches and upgrade overloaded switches.
- Establish additional Virtual Circuits to reduce congestion.
- Balance traffic across multiple ATM paths.
- Increase available bandwidth where necessary.

Quality of Service (QoS) Improvements

- Configure **CBR** for VoIP.
- Configure **VBR-RT (Variable Bit Rate - Real Time)** for video conferencing.
- Configure **ABR (Available Bit Rate)** for general data transfer.
- Configure **UBR (Unspecified Bit Rate)** for non-critical background traffic.

Traffic Engineering

- Prioritize mission-critical financial transactions.
- Implement traffic shaping and policing.
- Optimize routing to avoid congested ATM switches.

Performance Monitoring

- Continuously monitor:
 - ATM switch utilization
 - Cell loss ratio
 - End-to-end delay
 - Jitter
 - Bandwidth usage
 - Virtual Circuit performance

Reliability Improvements

- Provide redundant ATM links.
- Implement failover mechanisms.

- Regularly maintain ATM switches and firmware.

Expected Outcomes

- Faster online banking transactions.
- Reduced network delay and congestion.
- Smooth VoIP communication with minimal jitter.
- High-quality video conferencing.
- Efficient bandwidth utilization.
- Balanced traffic across ATM switches and Virtual Circuits.
- Improved Quality of Service (QoS).
- Reliable, secure, and uninterrupted enterprise communication.

3) Problems Identified (Root Causes)

- 1. Slow Access to Cloud Applications**
 - Congested Label Switched Paths (LSPs).
 - High network latency.
 - Improper bandwidth allocation.
- 2. Delayed Order Processing**
 - Incorrect label mappings in MPLS routers.
 - Congested backbone links.
 - Inefficient packet forwarding.
- 3. Poor VoIP Quality**
 - Lack of Quality of Service (QoS).
 - Network congestion causing jitter and packet loss.
 - Delay in voice packet delivery.
- 4. Video Conferencing Interruptions**
 - High latency and packet loss.
 - Insufficient bandwidth for real-time traffic.
 - Congested MPLS paths.
- 5. Congested Label Switched Paths (LSPs)**
 - Heavy traffic concentrated on a few LSPs.
 - Other available paths remain underutilized.
 - Poor traffic engineering configuration.
- 6. Incorrect Label Mappings**
 - Misconfigured MPLS routers.
 - Incorrect Label Distribution Protocol (LDP) configuration.
 - Routing errors increasing network delay.
- 7. Uneven Bandwidth Utilization**
 - Traffic Engineering (TE) not configured properly.
 - No load balancing across multiple paths.
 - Inefficient resource allocation.

Recommended Solutions

MPLS Network Optimization

- Correct label mappings on all MPLS routers.
- Verify and reconfigure Label Distribution Protocol (LDP).
- Establish optimized Label Switched Paths (LSPs).
- Upgrade congested network links where necessary.

Traffic Engineering (TE)

- Implement MPLS Traffic Engineering (MPLS-TE).
- Balance traffic across multiple LSPs.
- Optimize routing to avoid congested paths.
- Reserve bandwidth for critical business applications.

Quality of Service (QoS)

- Prioritize ERP, VoIP, and video conferencing traffic.
- Classify traffic based on application priority.

- Apply queuing and congestion management techniques.
- Reduce latency, jitter, and packet loss.

Performance Monitoring

- Continuously monitor:
 - LSP utilization
 - Router CPU and memory usage
 - Bandwidth utilization
 - Packet loss
 - Network latency
 - Label distribution status

Reliability Improvements

- Configure redundant MPLS links.
- Enable automatic failover and fast reroute.
- Perform regular router firmware and software updates.
- Conduct periodic network performance audits.

Expected Outcomes

- Faster access to cloud applications.
- Quicker online order processing.
- High-quality VoIP communication.
- Smooth and uninterrupted video conferencing.
- Balanced utilization of Label Switched Paths (LSPs).
- Reduced network latency and congestion.
- Efficient bandwidth usage through MPLS Traffic Engineering.
- Reliable, secure, and high-performance enterprise network communication.

4) Problems Identified (Root Causes)

- 1. Incorrect Shipment Records**
 - Corrupted data during transmission.
 - Undetected transmission errors.
 - Weak error detection mechanisms.
- 2. Barcode Scan Failures**
 - Signal interference affecting data transmission.
 - Damaged or incomplete data frames.
 - Temporary network congestion.
- 3. Inventory Mismatch**
 - Lost or corrupted packets.
 - Incorrect database updates.
 - Failure to detect all transmission errors.
- 4. Transmission Errors**
 - Electrical interference (noise).
 - Signal attenuation over long distances.
 - Network congestion causing frame corruption.
- 5. Undetected Corrupted Frames**
 - Existing error detection techniques are insufficient.
 - Some damaged frames pass integrity checks.
 - Missing error correction mechanisms.
- 6. Delayed Order Processing**
 - Frequent retransmissions.
 - Increased network latency.
 - Database inconsistencies caused by transmission errors.

Weaknesses in Existing Error Control

- Simple error detection methods may miss some errors.
- Lack of efficient error correction.
- Excessive retransmissions reduce network performance.

- No prioritization of critical warehouse data.
- Limited monitoring of transmission quality.

Recommended Solutions

Error Detection

- Implement **Cyclic Redundancy Check (CRC)** for accurate error detection.
- Use **Checksums** and **Parity Bits** for additional validation.
- Verify data integrity before updating the database.

Error Correction

- Apply **Forward Error Correction (FEC)** where possible.
- Use **Automatic Repeat reQuest (ARQ)** protocols to retransmit corrupted frames.
- Enable acknowledgements (ACK) and retransmission for lost frames.

\text{Learn more}

Network Optimization

- Reduce electrical interference using shielded cables and proper grounding.
- Upgrade network hardware to minimize signal attenuation.
- Increase available bandwidth during peak hours.
- Implement Quality of Service (QoS) to prioritize barcode and inventory traffic.

Performance Monitoring

- Continuously monitor:
 - Frame error rate
 - CRC errors
 - Packet loss
 - Network congestion
 - Signal strength
 - Retransmission rate

Database Reliability

- Validate received data before database updates.
- Maintain transaction logs and automatic backups.
- Synchronize warehouse inventory with the central server regularly.

Expected Outcomes

- Accurate shipment records.
- Reliable barcode scanning.
- Correct inventory database updates.
- Reduced transmission errors.
- Faster and more reliable data communication.
- Lower packet loss and retransmissions.
- Improved network reliability and warehouse efficiency.
- Faster order processing and higher customer satisfaction.

5) Problems Identified (Root Causes)

1. Incomplete Downloads

- Sender transmits data faster than receivers can process.
- Network congestion during peak hours.
- Packet loss causing incomplete file transfer.

2. Delayed Page Loading

- High network latency.
- Limited bandwidth for many simultaneous users.
- Excessive retransmissions.

3. Duplicate Packets

- Unnecessary retransmissions due to packet loss.
- Inefficient flow control mechanism.
- Delayed acknowledgements (ACKs).

4. Receiver Buffer Overflow

- Receiver cannot process incoming data quickly enough.

- Buffer memory becomes full.
- Application crashes due to overflow.
- 5. **Increased Retransmissions**
 - Packet loss caused by congestion.
 - Timeout before ACK is received.
 - Poor congestion and flow control.
- 6. **Reduced Throughput**
 - Frequent retransmissions consume bandwidth.
 - Congestion lowers effective data transfer rate.
 - Slow receiver devices reduce overall performance.
- 7. **Poor Bandwidth Utilization**
 - Uneven data transmission rates.
 - No proper sender rate control.
 - High-speed server overwhelms slower client connections.

Weaknesses in Existing Flow Control

- Sender transmits data without considering receiver capacity.
- Buffer management is inefficient.
- Flow control windows are not optimized.
- Congestion control is insufficient during peak traffic.
- Lack of adaptive transmission rate adjustment.

Recommended Solutions

Flow Control Optimization

- Implement **Sliding Window Protocol**.
- Use **TCP Flow Control** to match the sender's transmission rate with the receiver's processing speed.
- Dynamically adjust window size based on network conditions.

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Congestion Control

- Implement **TCP Congestion Control** (Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery).
- Limit transmission rate during congestion.
- Reduce unnecessary retransmissions.

Buffer Management

- Increase receiver buffer capacity where possible.
- Apply efficient buffer management to prevent overflow.
- Process received packets before accepting new data.

Bandwidth Optimization

- Use **Quality of Service (QoS)** to prioritize:
 - Online examination data
 - Question papers
 - Answer submissions
 - Live class video traffic
- Compress images and videos to reduce bandwidth usage.
- Use Content Delivery Networks (CDNs) for faster content distribution.

Performance Monitoring

- Continuously monitor:
 - Network latency
 - Throughput
 - Packet loss
 - Retransmission rate
 - Buffer utilization
 - Bandwidth usage

Expected Outcomes

- Complete and reliable downloads.

- Faster page loading during examinations.
- Elimination of receiver buffer overflow.
- Reduced packet retransmissions.
- Improved network throughput.
- Efficient bandwidth utilization.
- Stable online examinations and live classes.
- Better user experience for all students.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation